

# Mitsue Forest → Energy: A Workforce-Led 25-Year Plan

"The forest our ancestors planted — the power that sustains the village they built" 先人が植えた森が、今、村を支える力になる。

## 1. The core idea

The bottleneck in Mitsue's forest is **not the trees — it is the crew and the machines**. 御杖村森林組合 (Mitsue Kanko) today manages only a fraction of the village's **7,051 ha** of forest, most of it as subsidised thinning rather than full harvest.

This project's purpose is to **double — and over time triple — the cooperative's workforce and mechanise it** (harvester + forwarder, chipper, drying). Energy revenue from the harvested sugi pays for that crew and equipment; the larger crew harvests more; more harvest feeds a larger biomass plant. It is a **self-funding loop**, the same model proven in Shimokawa and Nishiawakura.

## 2. Where the cooperative is today

| Item                   | Current                                                   | Source       |
|------------------------|-----------------------------------------------------------|--------------|
| Village forest area    | ~7,051 ha (>90% private, sugi/hinoki postwar plantations) | Co-op report |
| Annual thinning        | ~66 ha/yr                                                 | Co-op report |
| Raw material recovered | only ~120–300 m³/yr                                       | Co-op report |
| Crew                   | small, young (avg age ~33)                                | Co-op report |

The recovered volume is tiny relative to the area worked because the current regime is **subsidised thinning, not mechanised harvest**. That is precisely the gap the project closes.

## 3. Real precedents we can check

| Village / Town                      | Forest                               | Harvest model                                                                           | Energy result                                              |
|-------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Shimokawa, Hokkaido</b> (下川町)    | 4,500+ ha town forest (FSC 7,150 ha) | <b>~50 ha/yr</b> harvest + <b>replant</b> , 60-yr sustained cycle                       | 56% local heat self-sufficiency; ~20% town CO <sub>2</sub> |
| <b>Nishiawakura, Okayama</b> (西粟倉村) | 93–95% forest                        | ~ <b>3,000 ha</b> under 100-yr management contracts (village manages on owners' behalf) | forestry-jobs revival                                      |
| <b>Maniwa, Okayama</b> (真庭市)        | city-scale                           | <b>148,000 t/yr</b> (90k unused wood + 58k mill residue)                                | <b>10 MW</b> , 79.2 GWh/yr, ~22,000 homes, 95% uptime      |
| <b>Mishima, Fukushima</b> (三島町)     | ~88% forest                          | ~750 t/yr                                                                               | ≤50 kWe — our model's calibration anchor                   |

**Shimokawa** is the closest sustained-cycle model: a whole town forest run on **~50 ha/yr by a professional, mechanised crew**. It shows what a scaled-up Mitsue Kanko can look like.

## 4. What the workforce can realistically deliver

Harvest volume — and therefore plant size — is set by how large and how mechanised the crew becomes. Figures below are from the project's calibrated **Forest Twin** model (sugi 600 m<sup>3</sup>/ha, 314 kg/m<sup>3</sup>, 18.5 GJ/t; net electrical efficiency 0.13, calibrated so a Mishima-scale unit reproduces ~50 kWe on ~750 t/yr). All harvested wood is routed to fuel; conversion runs over 25 years.

| Cooperative capacity                       | Harvest rate      | Area in 25 yr         | Wood / yr              | Biomass plant     | Electricity          | Heat              |
|--------------------------------------------|-------------------|-----------------------|------------------------|-------------------|----------------------|-------------------|
| Current + light mechanisation              | ~70 ha/yr         | 1,762 ha (25%)        | 10,500 m <sup>3</sup>  | ~290 kWe          | ~2,000 MWh/yr        | ~6,900 MWh-th/yr  |
| <b>Double crew + harvester/forwarder ✓</b> | <b>~140 ha/yr</b> | <b>3,526 ha (50%)</b> | 42,000 m <sup>3</sup>  | <b>~1,150 kWe</b> | <b>~8,000 MWh/yr</b> | ~27,700 MWh-th/yr |
| Triple crew, full mechanisation            | ~200 ha/yr        | ~5,000 ha (~70%)      | ~60,000 m <sup>3</sup> | ~1.6 MWe          | ~11,000 MWh/yr       | ~38,000 MWh-th/yr |
| <i>(Optimistic 90% — needs ~4 × crew)</i>  | 254 ha/yr         | 6,346 ha (90%)        | 135,000 m <sup>3</sup> | 3.7 MWe           | ~26,000 MWh/yr       | ~89,000 MWh-th/yr |

✓ **Recommended baseline: the doubled-workforce case — ~50% of the forest, ~1.1–1.2 MWe.**

## 5. Recommended plant

Build the plant as **2 × ~0.6 MWe units**, not one 1.2 MWe unit:

- **Redundancy / maintenance** — one unit can be serviced while the other runs.
- **FIT** — both stay under the 2 MW unused-material (未利用材) band, capturing the higher tariff (~¥40/kWh vs ~¥32/kWh for ≥2 MW).
- **Phased growth** — install unit 1 as the crew doubles, unit 2 as it approaches triple. The plant grows in step with the workforce.

Indicative capex at this scale ≈ ¥0.8 billion (¥700k/kWe installed). Electricity → grid or salt-battery thermal store; the recovered heat (~27,700 MWh-th/yr) also charges the store or supplies district heat.

## Reference equipment — makers (domestic only)

**Project requirement: the CHP unit must be a Japanese manufacturer** — for domestic service/parts, simpler FIT + 交付金 paperwork, and to reinforce the local supply-chain / endogenous-development story with the village and 御杖村森林組合. Small woody-gasification units mostly top out at ~200–500 kWe, so ~0.6 MWe is a **cluster of modules**.

| Maker                                 | Tech / scale                          | Fit           | Notes                                                                                    |
|---------------------------------------|---------------------------------------|---------------|------------------------------------------------------------------------------------------|
| 中外炉工業 (Chugai Ro)                     | rotary-kiln gasification cogeneration | ✓ <b>lead</b> | Since 1997, NEDO-backed; reaches our per-unit size; tolerates rough fuel (branches/bark) |
| 神鋼環境ソリューション (Shinko / Kobe Steel grp) | ~500 kWe-class up                     | backup        | △ confirm <b>gasification</b> vs steam (big plants are combustion)                       |

| Maker                 | Tech / scale                | Fit     | Notes                                                          |
|-----------------------|-----------------------------|---------|----------------------------------------------------------------|
| 静岡製機 (Shizuoka Seiki) | small ~200 kWe outdoor unit | cluster | METI-funded, aimed at agriculture/forestry — closest in spirit |
| ネオナイト (Neonite)       | downdraft two-stage         | cluster | Recovers charcoal + wood tar (biochar angle)                   |

The **JWBA list** (小規模木質バイオマス発電機器の一覧) catalogues all domestic units with specs and is the single best reference. Vendor/site visits are being pursued — see [mitsue\\_biomass\\_visit\\_request\\_emails.md](#).

**Imported units are benchmark-only, not for procurement.** FORTES Energy (Latvia; 0.14–1.2 MWe gasification CHP, new Tokyo entity), Spanner Re<sup>2</sup> and Burkhardt (Germany) match the size well and are useful for spec comparison, but do not meet the Japanese-manufacturer requirement.

## 6. Fuel handling & drying — equipment and cost

Site the plant at the existing **Mitsue Kanko processing centre** (牛峠工場, 神末797), which already chips and dries wood, so the CHP's heat feeds the dryers directly (no heat transport) and closes the loop **CHP heat → dry chips → CHP fuel**.

Fuel prep uses a two-stage regime: **(1) free roadside log seasoning (3–6 months, 55%→~35% moisture)**, then **(2) a short active polish-dry on CHP waste heat (~1–2 h, →10–15% for the gasifier)**. Drying load ≈ **9,000 MWh-th/yr**, about one third of the CHP's recovered heat.

Equipment and planning-grade cost (Japan, sized to ~13,000 dry-t/yr):

| Stage              | Equipment                                                         | Indicative cost (¥M)      |
|--------------------|-------------------------------------------------------------------|---------------------------|
| Roadside seasoning | Log covers/tarps, staging area                                    | 0.5–2                     |
| Chipping           | Drum/disc chipper (~10–20 t/h) ( <i>existing 牛峠工場 may cover</i> ) | 5–30                      |
| Passive store      | Ventilated covered chip barn (~500–1,000 m <sup>2</sup> )         | 15–40                     |
| Active polish-dry  | Belt / rotary drum dryer (~2–3 t/h water, CHP-heat fed)           | 30–80                     |
| Screening          | Sieve / screen deck                                               | 3–8                       |
| Handling           | Walking-floor bin, augers, conveyors                              | 10–25                     |
| Yard               | Wheel loader / telehandler                                        | 8–20                      |
| Instrumentation    | Moisture meters, controls, weighbridge                            | 3–10                      |
|                    | <b>Fuel-prep sub-total (excl. CHP gensets)</b>                    | <b>~75–215 (mid ~120)</b> |

CHP gensets (2×0.6 MWe) are a separate ~¥800M. **Two things cut the fuel-prep bill:** the existing 牛峠工場 kit may be sunk cost (–¥30–70M), and buying the dryer + screening

- handling **inside the gasifier vendor's package** (as Spanner Re<sup>2</sup>/Burkhardt supply in Japan) guarantees the heat/moisture match. These are scoping figures — firm numbers need a 牛峠工場 inventory audit and a vendor quote once the unit size is fixed.

**Budget placement:** this fuel-prep capex and the CHP fleet are **Phase 4 (Operations & Scale)** items, funded by operating revenue + grants — **outside** the Phase 0–3 EVM Performance Measurement Baseline (BAC ¥220M). A small pilot-scale drying line legitimately sits inside Phase 3 (WBS 5.6 forestry ops). See `mitsue_evm_plan.md` §14.

## 7. Specialty broadleaf revenue — the Tenkawa model

The 25-year plan converts sugi monoculture to native broadleaf (caveat §8.1). That replant does **not have to be a cost centre**: the neighbouring Yoshino village of **Tenkawa (天川村)**, ~70 km south, has turned the same conversion into a **second revenue stream** by replanting *marketable* broadleaf instead of generic forage species. This directly answers the "why replant when sugi doesn't pay" problem and is more palatable to conservation partners (e.g. more trees) than an energy-only harvest.

**Tenkawa Forest Power Council (天川村フォレストパワー協議会)** — formed Dec 2016 by the **village + forest cooperative + chamber of commerce** (the same three-body structure as 御杖村 + 御杖村森林組合). On clear-cut sugi land they replant high-value natives with existing markets:

| Species                                             | Product / market                                                                                         | To harvest |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------|
| キハダ <b>Kihada</b> ( <i>Phellodendron amurense</i> ) | bark for <i>Daranisuke</i> 陀羅尼助, a 1,300-yr-old herbal medicine — currently imported into the prefecture | ~20 yr     |
| ホオノキ <b>Hoonoki</b> ( <i>Magnolia obovata</i> )     | 朴葉 leaves for sushi wrapping                                                                             | —          |
| クロモジ <b>Kuromoji</b> ( <i>Lindera umbellata</i> )   | aroma / essential oil                                                                                    | —          |
| ウルシ <b>Urushi</b>                                   | lacquer                                                                                                  | —          |
| maple, mizunara oak, horse chestnut, katsura        | timber / craft / biodiversity                                                                            | —          |

**Why this is the same conversion pulse, monetised twice:** the sugi harvest funds the crew + CHP (\$4); the *replant* — instead of pure forage — is chosen for a downstream product, so the 5–20 yr gap before the new forest matures throws off **byproducts and craft income** rather than only cost. Kihada bark localises a supply Tenkawa currently buys out-of-prefecture.

**Enablers Tenkawa proves work here (same terrain, same deer pressure):**

- **Funding:** 農林水産業みらい基金 (Mirai Fund) grant covers nursery, forest roads and deer fencing — a template beyond the MoE 交付金 ladder.
- **Deer mitigation:** 12 m reinforced-net patch enclosures (3 m triangles failed) — directly reusable for Mitsue's replant.
- **Nursery + tending labour:** run as corporate/eco-tour volunteer days (more trees ran 育林・育苗 days there May 2024 with corporate volunteers) — an income + engagement channel that also feeds the school / eco-tourism programme.
- **Value-add:** partner Pony no Sato Farm runs bark-stripping, natural-dyeing and woodworking workshops and plans kihada-heartwood furniture — i.e. the products stay local.

**Training route for Mitsue Kanko:** Tenkawa also runs the 天川村森林塾 (**Tenkawa Forest School**) — an 8-day felling/chainsaw course that has trained **60+ people since 2017, explicitly "from inside AND outside the village."** It is the concrete peer-village channel for up-skilling the doubled Mitsue Kanko crew (§1). Contact: ten.forestpower@gmail.com / 0747-63-0321.

**Action:** open a knowledge-exchange with the Tenkawa Forest Power Council and 森林塾, and select **1-2 marketable broadleaf species** (kihada is the obvious lead) to fold into the replant spec so the conversion has a product, not just forage, behind it. The NGO still owns final species choice for wildlife (see reforestation principle).

## 8. Honest caveats

- 1. Sustainability depends on the regime.** One-way conversion (clearfell → broadleaf) is a finite fuel pulse; a **rotation / mixed** regime keeps the plant fed indefinitely. The 25-year plan should be framed as a deliberate *sugi-monoculture* → *native-forest conversion*, funded by the energy pulse, with replanting for wildlife forage and watershed protection.
- 2. Accessible area is unknown.** Steep roadless stands and fragmented private ownership (>90% private — Nishiawakura's and Mishima's #1 barrier) mean the "harvestable" fraction must be measured, not assumed. A drone + LiDAR survey (as used in Mishima) is the tool, and doubles as a trust-building gift to landowners.
- 3. The crew is the constraint, not the trees.** Every figure above scales with workforce and mechanisation — the project's central investment.

**Funding pathways for the Phase-4 CHP fleet (beyond the MoE 再エネ交付金 ladder):** Two FY2026 national subsidies fit the biomass-power-plus-compute model directly — 経産省 GX地域共創補助金 (¥210B pool; funds *decarbonized power* + DC/factory as one project; Round 1 Jul-Sep 2026) and 環境省 データセンターのゼロエミッション化・地域共生加速化事業 (≤¥1B/project on DC-linked renewables + storage; FY2026 window closed 2026-07-03 — aim for FY2027). Both are logged for Baseline Rev 2 in the EVM plan §12.

## Sources / 出典

- 御杖村森林組合 report (village forest ~7,051 ha; ~66 ha/yr thinning; ~120–300 m<sup>3</sup>/yr; crew avg age ~33) — Docs extra/御杖村森林組合 (Mitsue-mura Forest Owners' Cooperative) Report.md
- Ooba, Nakamura & Togawa (2020), *Promoting Local Revitalization to Solve Issues on Degraded Forests in Japan* — NIES Fukushima; Mishima Town CHP (≤50 kWe / 750 t/yr).
- Forest Twin model — Docs extra/CHP biomass/forest-twin/forest\_model.py (calibrated to Mishima; all assumptions inline).
- Shimokawa Town 循環型森林経営 — <https://www.town.shimokawa.hokkaido.jp/jigyo/2020/01/post-10.html>
- Shimokawa Town, MAFF interview — <https://www.maff.go.jp/j/shokusan/renewable/energy/interview/shimokawa.html>
- Nishiawakura Village 百年の森林構想 — <https://www.vill.nishiawakura.okayama.jp/wp/百年の森林構想/>
- Maniwa biomass (Renewable Energy Institute, 2017) — <https://www.renewable-ei.org/activities/column/20170620.html>
- Tenkawa Forest Power Council (キハダ specialty-broadleaf replant, Mirai Fund) — <https://www.miraikikin.org/activities/forestry/tenkawa.html>
- more trees 育林・育苗 in Tenkawa (May 2024, corporate volunteers, deer-net enclosures) — <https://www.more-trees.org/news/20240529/>
- 天川村森林塾 (Tenkawa Forest School; 60+ trainees since 2017, open to outsiders) — <https://tenforestpower.wixsite.com/tenkawajyuku>
- 経産省 GX地域共創補助金 (脱炭素電源地域貢献型投資促進事業; ¥210B, DC+電源一体支援) — <https://sustainablejapan.jp/2026/05/30/medi-gx-local/126036>
- 環境省 データセンターのゼロエミッション化・地域共生加速化事業 (≤¥1B/件; 締切 2026-07-03) — <https://sustainablejapan.jp/2026/06/07/japan-datacenter-renewable-energy/126312>